Primary CNS lymphoma

Potentially curable — avoid steroids before biopsy, a high-dose methotrexate backbone, and thiotepa-based transplant consolidation for the fit. Key trade-off: **cure versus neurotoxicity**

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — primary CNS and HIV-associated lymphoma · NCCN · IPCG · primary trial reports verified against PubMed, September 2026.

WHAT IT IS

- **DLBCL confined to brain, leptomeninges, spinal cord or eyes**, no systemic disease. Almost all are **non-GCB** / **ABC-type**, with **MYD88 L265P and CD79B** mutations — the reason BTK inhibitors and lenalidomide work
- WHO-HAEM5 groups it as **primary large B-cell lymphoma of immune-privileged sites** (with vitreoretinal and testicular); ICC keeps **primary DLBCL of the CNS**
- Median age late 60s. **Immunodeficiency (HIV, transplant) → EBV-driven** disease, often multifocal and ring-enhancing
- Classic MRI: **homogeneously enhancing periventricular / deep lesion**, restricted diffusion. Presents with focal deficits, personality change, raised ICP

DIAGNOSIS

- **AVOID CORTICOSTEROIDS BEFORE BIOPSY** unless herniation is imminent — steroids lyse lymphoma cells and the lesion can vanish ("ghost tumor"), leaving a non-diagnostic biopsy. If already given and biopsy is negative, stop them and re-image
- **Stereotactic needle biopsy** is the procedure — resection adds nothing and is not the goal
- Tissue substitutes if biopsy is unsafe: **CSF cytology + flow cytometry** (and MYD88 L265P) or **vitrectomy** for vitreous involvement

STAGING — IPCG EXTENT-OF-DISEASE WORKUP

Brain	MRI with gadolinium (+ spine MRI if symptoms or CSF+)
Eyes	**Slit-lamp + dilated fundus exam** by ophthalmology — vitreoretinal disease changes local therapy and follow-up
CSF	LP if safe (no mass effect): **cytology, flow, protein** (protein feeds the IELSG score)
Exclude systemic	**PET/CT** (or CT C/A/P) · **testicular exam ± ultrasound in men** · marrow biopsy selectively
Labs	**HIV**, HBV/HCV, LDH, **creatinine clearance** (MTX eligibility), albumin, and look for effusions/ascites

TREATMENT PATHWAY

Fit, transplant-eligible (to ~70)

↓

Induction: HD-MTX backbone ≥3 g/m²
MATRix ×4 (MTX + cytarabine + thiotepa + R)
or R-MPV ×5–7, or MT-R (CALGB 50202)
MRI after every 2 cycles

↓

Consolidation if CR / PR
thiotepa-based HDC-ASCT (IELSG43: PFS + OS)
not ASCT-fit: EA / HD-cytarabine, or
reduced-dose WBRT 23.4 Gy (after R-MPV CR)

Older / unfit for MATRix

↓

HD-MTX-based, gentler partners
R-MPV, MT-R, R-MP, or **HD-MTX + R**
fit >65 years: ASCT on trial (PRIMA-CNS)

↓

Then maintain
procarbazine, temozolomide or **lenalidomide**
avoid WBRT — neurotoxicity worst in elderly

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MTX not possible
rituximab + temozolomide · WBRT · BTKi
best supportive care with steroids

Relapsed / refractory

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Clinical trial first
late relapse (>12 mo): **re-challenge HD-MTX**
ibrutinib or lenalidomide ± R; HD-cytarabine
ASCT if chemosensitive and not done · WBRT
CAR-T on trial · tirabrutinib (PDUFA Dec 2026)

PROGNOSTIC SCORES

IELSG *Ferreri, JCO 2003*	1 point each: **age >60** · **ECOG >1** · **LDH** ↑ · **CSF protein** ↑ · **deep brain** (periventricular, basal ganglia, brainstem, cerebellum). 2-yr OS **0–1: 80%** · 2–3: 48% · **4–5: 15%**
MSKCC *Abrey, JCO 2006*	Class 1 **age <50** · class 2 ≥50 + KPS ≥70 · class 3 ≥50 + KPS <70 — survival falls stepwise; needs only age and KPS

CONSOLIDATION — ASCT VS WBRT

Trial	Result
IELSG32 2nd rand. *Lancet Haem 2017*	WBRT vs thiotepa-ASCT after the induction randomization: 2-yr PFS **80% vs 69%, P=0.17** — equivalent disease control, but **cognitive decline after WBRT**
PRECIS *Houillier, JCO 2019*	Age ≤60, randomized phase 2: 2-yr PFS **63% WBRT vs 87% ASCT**; cognition impaired after WBRT, preserved after ASCT; toxic deaths 1 vs 5
Alliance 51101 *Batchelor, Blood Adv 2024*	MT-R-A then thiotepa/BCNU-ASCT vs EA: 2-yr PFS **73% vs 51%** (P=0.02), but 28% of the EA arm progressed/died in induction; landmark from consolidation 86% vs 71%, NS
IELSG43 *Illerhaus, Lancet 2026*	**Phase 3**, n=229, MATRix ×4 then ASCT vs R-DeVIC: 3-yr PFS **78% vs 51%**, HR 0.43; **OS HR 0.46** (5-yr 84% vs 63%). Fatal events 5 vs 2 — **settles ASCT as the standard for fit patients**

WBRT remains effective but the late leukoencephalopathy, ataxia and dementia make it the last choice for anyone likely to be cured — if used after CR, **reduced-dose 23.4 Gy** (R-MPV protocol).

ELDERLY AND UNFIT

ANOCEF-GOELAMS *Omuro, Lancet Haem 2015*	≥60 yr, randomized phase 2 (non-comparative): MPV-A vs MTX-temozolomide — 1-yr PFS **36% in both**; median OS 31 vs 14 mo; **no late neurotoxicity** without WBRT
PRIMAIN *Fritsch, Leukemia 2017*	≥65 yr, R-MP(L) + **procarbazine maintenance**: CR 36%, 2-yr PFS 37%, OS 47%
US real-world *David, AJH 2023*	539 patients ≥60: combination HD-MTX beat MTX ± R (3-yr PFS 55% with MT-R vs 30%); **maintenance** 3-yr PFS 65% vs 45% — retrospective

RELAPSED / REFRACTORY AGENTS

Ibrutinib 560 mg	**iLOC** phase 2 (n=52): response 52% ITT (CR 19%) at 2 mo, median PFS **4.8 mo**; reaches CSF. **Invasive aspergillosis** reported (1 fatal). MSKCC phase 1: 10/13 responded. **Off-label in the US**
Tirabrutinib	PROSPECT phase 2 (n=48): ORR 67%, CR 44%, median PFS 6.0 mo (ASCO 2025 abstract). Approved in Japan; **US NDA under accelerated review, PDUFA 18 Dec 2026** — would be the first US-approved BTKi for PCNSL
Lenalidomide ± R	**REVRI** phase 2 (n=50): ORR 36% at end of induction, median PFS **7.8 mo**. Lenalidomide maintenance after salvage prolonged response (UCSF, Blood Adv 2018)
Re-challenge HD-MTX	Reasonable if the prior response lasted >12 months

R/R responses are frequent but short — **consolidate a second remission** (ASCT if not yet done, or maintenance) wherever possible.

Primary CNS lymphoma

REGIMEN DOSING (MG/M² UNLESS STATED)

Regimen	Dosing
MATRix *IELSG32/43*	Rituximab 375 d–5 and d0 · **MTX 3500 d1** (over 3–4 h) · cytarabine 2000 q12h ×4 d2–3 · **thiotepa 30 d4** — q21d ×4, G-CSF. Age 18–70
R-MPV *Shah, JCO 2007*	**Rituximab 500** d1 · MTX 3500 + vincristine 1.4 (handbook cap 2.8 mg) d2 · procarbazine 100 PO d1–7 odd cycles (low-tyramine diet) — q14d ×5; PR → 7 cycles. CR → WBRT 23.4 Gy, then cytarabine 3000 d1–2 ×2
MT-R *CALGB 50202*	**MTX 8000 over 4 h** d1 q14d ×8 · rituximab 375 ×6 doses · temozolomide 150 PO d7–11 every other cycle. Then **EA**: etoposide 40 mg/kg CIV over 96 h + cytarabine 2000 q12h ×8 d1–4
HD-MTX + R	MTX 3500–8000 over 4 h q14d + rituximab 375 — elderly / frail backbone
HDC-ASCT	**Thiotepa-based** conditioning: carmustine + thiotepa (IELSG32/43, Alliance 51101) or thiotepa + busulfan + cyclophosphamide (PRECIS, MSKCC)
Maintenance	Lenalidomide 15 mg daily (handbook) · procarbazine 100 mg/day ×5 d q4wk (PRIMAIN) · temozolomide 150 d1–5 q28d (Reni)
MTX-ineligible	Rituximab 375 + temozolomide (handbook) · temozolomide alone 150 d1–5 q28d · WBRT

MT-R and EA dosing per CALGB 50202 (handbook lists EA etoposide as 10 mg/kg/day CIV d1–4, i.e. the same 40 mg/kg total). Prednisolone 1% eye drops with every HD-cytarabine.

SUPPORTIVE CARE

- **After the biopsy, dexamethasone freely** for edema, then taper; **PJP prophylaxis** on prolonged steroids (atovaquone around MTX)
- **Antiseizure drugs only if seizures** — no prophylaxis; prefer non-enzyme-inducing agents
- **VTE is common** in PCNSL; weigh anticoagulation against the post-biopsy bleeding window
- **Neurocognitive baseline** and follow-up; MRI q3 mo ×2 yr, then less often (late relapses happen)

HIGH-DOSE METHOTREXATE — ADMINISTRATION

Eligibility	**CrCl ≥50 mL/min** for full dose (50% if 10–50; omit <10). **No third-space fluid** — drain or avoid with **pleural effusion, ascites, edema** (MTX sequesters and leaks back, prolonging exposure)
Stop interacting drugs	**PPIs** · **TMP-SMX** · **NSAIDs** / **salicylates** · **penicillins** · **probenecid** (and levetiracetam has case reports) — hold PJP TMP-SMX until the level clears
Hydrate + ALKALINIZE	Half-normal saline + 100 mEq $NaHCO_3$/L at 500 mL/h ×2 h, then 125 mL/h. **Do not start until urine pH ≥7** and UOP >100 mL/h; **keep pH ≥7 until level <0.05–0.1 µM** (MTX precipitates in acid urine). Rescue: $NaHCO_3$ bolus or acetazolamide
Leucovorin	PCNSL protocols start **24 h after MTX start** (25 mg IV q6h, or 100 mg/m² after 8 g/m² MT-R) — vs 36 h after the 24-h Burkitt infusion. Continue until MTX <0.05 µM
Levels	Daily from 24 h, with Cr and UOP. **Escalate leucovorin** (100 mg/m² q6h) for a Cr rise >50% or levels above the protocol curve — handbook thresholds (24-h infusion): >20 µM at 24 h, >1 at 48 h, >0.1 at 72 h; **4-h PCNSL infusions run on their own nomogram**
Glucarpidase	**50 U/kg IV** once for level >1 µM with delayed clearance from renal failure. **No leucovorin within 2 h** either side; **immunoassay unreliable for 48 h** (DAMPA cross-reacts) — use chromatographic assay
Watch	AKI · transaminitis (common, transient) · mucositis · myelosuppression · **acute** / **subacute encephalopathy** (stroke-like; consider aminophylline, dextromethorphan)

SPECIAL SITUATIONS

- **HIV-associated PCNSL** is EBV-driven and arises with advanced immunosuppression. Differential is **toxoplasmosis** — toxo serology, **CSF EBV PCR**, and biopsy if lesions fail to shrink on empiric anti-toxoplasma therapy. Treat with **ART + HD-MTX-based therapy** with curative intent (handbook: continue or start ART during chemotherapy)
- **Vitreoretinal involvement:** systemic HD-MTX reaches the eye inconsistently — add **intravitreal MTX or rituximab, or ocular RT** if it persists (handbook: orbital RT or ophthalmology referral for intra-ocular therapy)
- **Response (IPCG):** CR requires no enhancing disease, **off steroids ≥2 weeks**, a normal eye exam and negative CSF cytology — steroid-driven shrinkage is not a response

KEY TRIALS

IELSG20 *Ferreri, Lancet 2009*	MTX ± HD-cytarabine: CR **46% vs 18%** — established the doublet
IELSG32 / MATRix *Lancet Haem 2016*	+ rituximab + thiotepa: CR **49% vs 23%**; 7-yr OS 56% vs 21%; 6% toxic deaths. Randomized phase 2
CALGB 50202 *Rubenstein, JCO 2013*	MT-R → EA, no WBRT, single arm: CR 66%, 2-yr PFS 57%
IELSG43 *Illerhaus, Lancet 2026*	ASCT vs R-DeVIC after MATRix: 3-yr PFS **78% vs 51%**; OS HR 0.46. Phase 3
PRECIS · Alliance 51101	ASCT ≥ WBRT or EA for disease control; WBRT costs cognition
HOVON 105 / ALLG *Bromberg, Lancet Oncol 2019*	MBVP ± rituximab, phase 3: **no EFS benefit** (HR 1.00)
ANOCEF elderly · PRIMAIN	Outcomes ≥60–65 remain poor: 1–2 yr PFS ~35%
iLOC · REVRI · PROSPECT	Ibrutinib, R2 and tirabrutinib in R/R disease

PEARLS

- **Steroids before biopsy are the #1 avoidable diagnostic failure**
- **Urine alkalinization is mandatory for HD-MTX** (the opposite of TLS, where it is harmful)
- **Intrathecal chemotherapy is generally omitted** when systemic MTX ≥3 g/m² is given — no benefit shown in retrospective series (R-MPV adds Ommaya MTX only for positive CSF)
- **Testicular exam in every man** — testicular DLBCL can seed the CNS and changes the diagnosis
- **Rituximab is contested:** it raised CR in IELSG32, but **HOVON 105/ALLG NHL 24 (phase 3, n=199) found no EFS benefit** (1-yr EFS 52% vs 49%, HR 1.00) — it stays in most regimens anyway